

Cedric Orton-Urbina

Cornell CS Master Graduate and Published Aerospace Researcher
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EDUCATION

- **Cornell University** Ithaca, New York
Masters of Engineering - Computer Science; GPA: 3.53 May 2026
Relevant Coursework: LLM Transformers; Robot Learning; Machine Learning; Software Testing; Software Engineering
- **Cornell University** Ithaca, New York
Bachelor of Arts - Physics, Concentration in Astrophysics; GPA: 3.58 May 2025
Relevant Coursework: Analytical Mechanics; Numerical Analysis and Optimization; Planetary Physics; Object-Oriented Programming

TECHNICAL PROJECTS

- **Zulip Follow User Feature** Python, Claude, TypeScript, PostgreSQL, REST APIs
Open-source, Client-Led Contribution Spring 2026
 - Designed a full-stack feature for the Zulip open-source communication platform.
 - Developed REST APIs, relational data models, caching, real-time events, and frontend state management.
 - Refined AI-generated, automated, and regression tests covering API behavior, database state, caching, events, and edge cases.
- **Cooperative Multi-Agent Reinforcement Learning** Python, EPyMARL, SMAC, Google Cloud
Machine Learning Research Project Spring 2026
 - Designed a study comparing QMIX / MAPPO, recurrent / feed-forward policies, and dense / sparse rewards.
 - Configured and executed GPU training runs totaling 6.3 million environment steps across random seeds.
 - Evaluated learning stability and reward sensitivity.
- **State Transition Matrix Integrator** Python, SciPy, SymPy, NumPy
Nonlinear Dynamics Library 2022 - 2024
 - Developed an open-source Python library for integrating and computing state transition matrices and tensors.
 - Built a pipeline that converts symbolic equations of motion into numerical state and sensitivity solvers.
 - Utilized the library to simulate and model spacecraft rendezvous maneuvers in support of a peer-reviewed publication.

PUBLICATIONS

- [1] J. Kulik, M. Ruth, **C. Orton-Urbina**, and D. Savransky. “Applications of Induced Tensor Norms to Guidance Navigation and Control”. In: *Journal of Guidance, Control, and Dynamics* 0.0 (0), pp. 1–19. DOI: [10.2514/1.G009054](https://doi.org/10.2514/1.G009054). eprint: <https://doi.org/10.2514/1.G009054>. URL: <https://doi.org/10.2514/1.G009054>.

RESEARCH EXPERIENCE

- **Cassini Data Analysis Program** Cornell University
Graduate Researcher for NASA Project Spring 2025 - Spring 2026
 - Wavelet and Fourier analysis of Saturnian ring data from the Cassini space probe.
 - Construction of ring wave models using windowed least squares regression and convolutional neural networks.
 - Usage of known orbital resonances to identify probable sources of oscillatory vertical structure.
- **Space Imaging and Optical Systems Lab** Cornell University
Undergraduate Researcher for Aerospace Applications Fall 2022 - Spring 2024
 - Researched nonlinear dynamics, orbital mechanics, flow maps, and higher-order sensitivity analysis.
 - Applied variational equations, state transition tensors, and induced tensor norms to trajectory perturbation.
 - Developed numerical simulations in two-body and restricted three-body dynamics for publication.

SKILLS

COMPUTATIONAL

- **Languages**

Proficient in C/C++, Java, JavaScript, TypeScript, Python, SQL, L^AT_EX

- **Scientific Computing and Flight Dynamics**

Experience with the following:

- Six degree of freedom spacecraft modeling and simulation with rendezvous and impulse transfers.
- Numerical integration of differential equations; trajectory propagation; state and sensitivity propagation.
- Two-body and restricted three-body mechanics; variational equations; state transition matrices and tensors.
- Numerical, computational, and symbolic linear algebra; Fourier and wavelet analysis; least-squares regression.

- **Software Development**

Experience with the following:

- Software Architecture: Parallel Processing; Concurrent Programming; Object Oriented Programming; Functional Programming; Containerization (Docker); Relational Databases (PostgreSQL).
- Software Systems: REST APIs; Event-Driven Architecture; Real-Time Client-Server Synchronization; Caching and Cache Invalidation.
- Software Testing: Automated; API; Mutation; Regression; Model Driven Test Design; Edge-Case Testing; Test Racks.
- Software Engineering: Process Model Selection; UML; Architectural Frameworks; Deployment and Infrastructure; Git Version Control; Debugging; Technical Documentation.

- **Optimization and Nonlinear Programming**

Experience with the following:

- Stochastic Gradient Descent, RMSprop, AdaGrad, and ADAM.
- Modified Newton, BFGS Quasi-Newton, and Gauss-Newton.
- Hyperparameter optimization via Wolfe, Lipschitz, and Armijo-stochastic line search.
- Nonlinear optimization with KKT analysis.
- Constrained Optimization using Nullspace, Barrier, Projected Gradient, and Stochastic methods.

ACADEMIC HONORS

- Cornell Dean's List for Excellence in Scholarship

Fall 2021; Spring 2022; Fall 2022; Fall 2024

LEADERSHIP AND SERVICE

- **President**

Seal and Serpent Society

Ithaca, NY

2024

- Worked with and managed executive board to provide enriching experiences to society members.
- Served as primary liaison between the Seal and Serpent Society and Cornell University.
- Served as primary interface between 119 years of alumni and active members.